

1. Evaluate $\int_0^1 x(\ln x)^2\,dx$.

   Give the answer as an integer or a fraction in lowest terms.

$\frac{1}{4}$

First note: $\int_0^1 x(\ln x)^2\,dx$ is improper integral bec. $x(\ln x)^2 \longrightarrow 0\cdot(-\infty)^2$ (?)

$$= \lim_{\varepsilon \to 0+} \int_\varepsilon^1 x(\ln x)^2\,dx$$

Compute anti-derivative $\int x(\ln x)^2\,dx$

$$= \left(\tfrac{1}{2}x^2\right)(\ln x)^2 - \int \left(\tfrac{1}{2}x^2\right)\cdot 2(\ln x)\tfrac{1}{x}\,dx$$

$\int$ by parts

$$= \left(\tfrac{1}{2}x^2\right)(\ln x)^2 - \int x(\ln x)\,dx$$

$$= \left(\tfrac{1}{2}x^2\right)(\ln x)^2 - \left[\left(\tfrac{1}{2}x^2\right)(\ln x) - \int \left(\tfrac{1}{2}x^2\right)\tfrac{1}{x}\,dx\right]$$

$\int$ by parts

$$= \left(\tfrac{1}{2}x^2\right)(\ln x)^2 - \left(\tfrac{1}{2}x^2\right)(\ln x) + \tfrac{1}{4}x^2 + \text{const}$$

so for any $\varepsilon > 0$,

$$\int_\varepsilon^1 x(\ln x)^2\,dx = \left[\left(\tfrac{1}{2}x^2\right)(\ln x)^2 - \left(\tfrac{1}{2}x^2\right)(\ln x) + \tfrac{1}{4}x^2\right]_{x=\varepsilon}^{x=1}$$

$$= 0 - 0 + \frac{1}{4} - \frac{\varepsilon^2}{2}(\ln\varepsilon)^2 + \frac{\varepsilon^2}{2}(\ln\varepsilon) - \frac{\varepsilon^2}{4}$$

so $$\int_0^1 x(\ln x)^2\,dx = \lim_{\varepsilon\to 0+}\int_\varepsilon^1 x(\ln x)^2\,dx = \lim_{\varepsilon\to 0+}\left(\frac{1}{4} - \frac{\varepsilon^2}{2}(\ln\varepsilon)^2 + \frac{\varepsilon^2}{2}(\ln\varepsilon) - \frac{\varepsilon^2}{4}\right)$$

$$= \frac{1}{4} - \lim_{\varepsilon\to 0+}\frac{\varepsilon^2}{2}(\ln\varepsilon)^2 + \lim_{\varepsilon\to 0+}\frac{\varepsilon^2}{2}(\ln\varepsilon) - 0 = \frac{1}{4}$$

$$= \frac{1}{2}\lim_{\varepsilon\to 0+}\frac{(\ln\varepsilon)^2}{\left(\frac{1}{\varepsilon}\right)^2} \qquad = \frac{1}{2}\lim_{\varepsilon\to 0+}\frac{(\ln\varepsilon)}{\left(\frac{1}{\varepsilon}\right)^2}$$

l'Hôpital's rule $= \frac{1}{2}\lim_{\varepsilon\to 0+}\frac{2(\ln\varepsilon)\cdot\frac{1}{\varepsilon}}{-2\left(\frac{1}{\varepsilon}\right)^3}$ $= \frac{1}{2}\lim_{\varepsilon\to 0+}\frac{\frac{1}{\varepsilon}}{-2\left(\frac{1}{\varepsilon}\right)^3}$ l'Hôpital's rule

$$= -\frac{1}{2}\lim_{\varepsilon\to 0+}\frac{(\ln\varepsilon)}{\left(\frac{1}{\varepsilon}\right)^2} = -\frac{1}{4}\lim_{\varepsilon\to 0+}\varepsilon^2 = 0$$

l'Hôpital's rule $= -\frac{1}{2}\lim_{\varepsilon\to 0+}\frac{\frac{1}{\varepsilon}}{-2\left(\frac{1}{\varepsilon}\right)^3}$

$$= \frac{1}{4}\lim_{\varepsilon\to 0+}\varepsilon^2 = 0$$

2. Evaluate $\lim_{x\to 2} \dfrac{(x-2)(\ln x - \ln 2)}{x^2 - 5x + 6 + 2\ln x - \ln 4}$.

Give the answer as an integer or a fraction in lowest terms.

$\frac{2}{3}$

$$\lim_{x\to 2} \frac{(x-2)(\ln x - \ln 2)}{x^2 - 5x + 6 + 2\ln x - \ln 4}$$

$$= \lim_{x\to 2} \frac{1\cdot(\ln x - \ln 2) + (x-2)\cdot\frac{1}{x}}{2x - 5 + \frac{2}{x}} \qquad (= 1 - \tfrac{2}{x})$$

(l'Hôpital's rule)

$$= \lim_{x\to 2} \frac{\frac{1}{x} + \frac{2}{x^2}}{2 - \frac{2}{x^2}}$$

(l'Hôpital's rule)

$$= \frac{\frac{1}{2} + \frac{2}{4}}{2 - \frac{2}{4}} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$$

3. Consider the function f given by

$$f(x) = \int_{x-2}^{x^2-7} \frac{1}{t^3+3}\,dt.$$

$\boxed{\frac{13}{44}}$

Find $f'(3)$.

Give the answer as an integer or a fraction in lowest terms.

FTC: says if $g(a) := \int_0^a \frac{1}{t^3+3}\,dt$ then $g'(a) = \left.\frac{1}{t^3+3}\right|_{t=a} = \frac{1}{a^3+3}$

so $f(x) = g(x^2-7) - g(x-2)$

so $f'(x) = g'(x^2-7)\cdot(2x) - g'(x-2)\cdot(1)$

$$= \frac{2x}{(x^2-7)^3+3} - \frac{1}{(x-2)^3+3}$$

so $f'(3) = \frac{2\cdot 3}{(3^2-7)^3+3} - \frac{1}{(3-2)^3+3} = \frac{6}{11} - \frac{1}{4} = \frac{13}{44}$ #

4. Let $P = (3, 2, 1)$ and $Q = (0, 1, 2)$. Find the point of intersection between the plane $2x + 3y + 6z = 14$ and the line through P, Q.

Give the answer in the form (a, b, c) where a, b, c are real numbers.

$\left(-1, \frac{2}{3}, \frac{7}{3}\right)$

Line thru PQ has eqn $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + t \cdot \begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix}$

P $\quad \overrightarrow{PQ} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix}$

Plane has eqn $2x + 3y + 6z = 14$

At intersectn point, (value of t is such that)

$x = 3 - 3t$
$y = 2 - t$
$z = 1 + t$
$2x + 3y + 6z = 14$

so $2(3 - 3t) + 3(2 - t) + 6(1 + t) = 14$

so $-3t = -4$ $\quad$ so $t = \frac{4}{3}$

so $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + \underbrace{t}_{\frac{4}{3}} \cdot \begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -4 \\ -4/3 \\ 4/3 \end{pmatrix}$

$= \begin{pmatrix} -1 \\ 2/3 \\ 7/3 \end{pmatrix}$

5. Let S be the surface in the xyz-space defined by the equation $3x^2 - 2y + 2z^2 = 4$. Find the point on the surface S at which the tangent plane to S is parallel to the plane $6x - y + 2z = 5$.

Give the answer in the form (a, b, c), where a, b, c real numbers.

$(2, 5, 1)$

Given plane has eqn $6x - y + 2z = 5$, has normal vector $\begin{pmatrix} 6 \\ -1 \\ 2 \end{pmatrix}$

S has eqn: $\underbrace{3x^2 - 2y + 2z^2}_{f(x,y,z)} = 4$, is a level surface of the fn $f(x,y,z)$

$\Rightarrow \nabla f(x,y,z) = \begin{pmatrix} 6x \\ -2 \\ 4z \end{pmatrix}$ is $\perp$ to tangent plane to S at point $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

gradient vector of f

so for this tangent plane to be parallel to given plane

require $\begin{pmatrix} 6x \\ -2 \\ 4z \end{pmatrix}$ to be a scalar multiple of $\begin{pmatrix} 6 \\ -1 \\ 2 \end{pmatrix}$

i.e. want x, y, z s.t. there exists $\lambda \in \mathbb{R}$ s.t.

$$\begin{pmatrix} 6x \\ -2 \\ 4z \end{pmatrix} = \lambda \begin{pmatrix} 6 \\ -1 \\ 2 \end{pmatrix} \quad \Rightarrow \quad \lambda = 2, \; x = 2, \; z = 1$$

and $3x^2 - 2y + 2z^2 = 4$ $\longrightarrow$ $3(2)^2 - 2y + 2(1)^2 = 4$

$\Rightarrow$ $y = 5$

6. Find the function $f(x,y)$ such that

$$\frac{\partial f}{\partial x} = \cos x + 4x^3y^2, \quad \frac{\partial f}{\partial y} = 2x^4y - \sin y,$$

and $f(0,0) = 7$.

Give the answer as an expression in terms of x and y.

$\sin x + x^4y^2 + \cos y + 6$

From $\frac{\partial f}{\partial x} = \cos x + 4x^3y^2$, get (anti-derivative wrt x)

$f(x,y) = \sin x + x^4y^2 + g(y)$ for some fn $g(y)$ of y

Partial diff wrt y, we get

$\frac{\partial f}{\partial y}(x,y) = 2x^4y + g'(y)$ (where $g'(y) = \frac{dg}{dy}$)

compared with given info that $\frac{\partial f}{\partial y} = 2x^4y - \sin y$

we deduce $g'(y) = -\sin y$

so $g(y) = \cos y + \text{constant}$

Hence $f(x,y) = \sin x + x^4y^2 + \cos y + \text{constant}$

so $f(0,0) = 0 + 0^4 0^2 + 1 + \text{constant}$, compared with given info that $f(0,0) = 7$

we deduce constant $= 6$

Hence $f(x,y) = \sin x + x^4y^2 + \cos y + 6$

7. Let $u = 3xy$, $v = v(x,y)$ be such that $\dfrac{\partial v}{\partial x} = 2x$ and $\dfrac{\partial v}{\partial y} = -2y$, $w = x^2 + xy$ and $z = u^2 + 2v + w^2$.

Determine $\dfrac{\partial z}{\partial y}$ as a function of x, y.

$$2x^3 + 20x^2y - 4y$$

$z = z(u,v,w)$

$(x,y) \in \mathbb{R}^2 \longrightarrow (u,v,w) \in \mathbb{R}^3 \longrightarrow z \in \mathbb{R}$

$u = 3xy$
$v = v(x,y)$
$w = x^2 + xy$

$\dfrac{\partial u}{\partial y} = 3x$
$\dfrac{\partial w}{\partial y} = x$

$z = u^2 + 2v + w^2$

$\dfrac{\partial z}{\partial u} = 2u$
$\dfrac{\partial z}{\partial v} = 2$
$\dfrac{\partial z}{\partial w} = 2w$

Chain rule :

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y} + \frac{\partial z}{\partial w}\cdot\frac{\partial w}{\partial y}$$

$$= 2u \cdot 3x + 2 \cdot (-2y) + 2w \cdot x$$

$$= 6xy \cdot 3x - 4y + (2x^2 + 2xy)\,x$$

$$= 18x^2y - 4y + 2x^3 + 2x^2y$$

$$= 2x^3 + 20x^2y - 4y$$

8. Consider the function $f(x,y) = x^2y - y^2$. Find the directional derivative of $f(x,y)$ at the point $(3,2)$ in the direction of the vector $8\mathbf{i} + 15\mathbf{j}$.

Give the answer as an integer or a fraction in lowest terms.

$\frac{171}{17}$

$$f(x,y) = x^2y - y^2 \quad \Rightarrow \quad \nabla f(x,y) = \begin{pmatrix} 2xy \\ x^2 - 2y \end{pmatrix}$$

So directional derivative of $f(x,y)$ at the point $(3,2)$ in the direction of the vector $8\mathbf{i} + 15\mathbf{j}$

$$= \nabla f(x,y)\Big|_{(3,2)} \cdot \frac{8\mathbf{i} + 15\mathbf{j}}{\|8\mathbf{i} + 15\mathbf{j}\|}$$

$$= \begin{pmatrix} 12 \\ 5 \end{pmatrix} \cdot \frac{\begin{pmatrix} 8 \\ 15 \end{pmatrix}}{\sqrt{8^2 + 15^2}} = \frac{1}{17}\begin{pmatrix} 12 \times 8 \\ + 5 \times 15 \end{pmatrix}$$

$$= \frac{171}{17}$$

9. Let $P(x,y,z)$, measured in pascals, be the air pressure at any point (x,y,z) in the xyz-space. Here the coordinates x, y, z are measured in meters. It is given that $\nabla P(x,y,z) = \langle yz - 3x^2,\ xz,\ xy - 6\rangle$. Suppose the position vector $\mathbf{r}(t)$ of a bird at time t, measured in seconds, is given by $\mathbf{r}(t) = \langle t^2,\ t^3,\ 2t+1\rangle$. Let A be a real number such that the rate of change of air pressure experienced by the bird at time $t = 1$ is A pascals per second. Find the value of A.

Give the answer as an integer or a fraction in lowest terms.

- 1

$t \in \mathbb{R} \xrightarrow{\mathbf{r}} (x,y,z) \in \mathbb{R}^3 \xrightarrow{P = P(x,y,z)} P \in \mathbb{R}$

$\mathbf{r}(t) = \langle t^2, t^3, 2t+1\rangle$

so $P(\mathbf{r}(t))$ is air press. experienced by bird

Want: $\frac{d}{dt} P(\mathbf{r}(t))$

Chain rule: $\frac{d}{dt} P(\mathbf{r}(t)) = \nabla P(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt}\Big|_t = \begin{pmatrix} yz - 3x^2 \\ xz \\ xy - 6 \end{pmatrix}\Big|_{(x,y,z) = t^2,\ t^3,\ 2t+1} \cdot \begin{pmatrix} 2t \\ 3t^2 \\ 2 \end{pmatrix}$

$$= \begin{pmatrix} t^3(2t+1) - 3t^4 \\ 2t^3 + t^2 \\ t^5 - 6 \end{pmatrix} \cdot \begin{pmatrix} 2t \\ 3t^2 \\ 2 \end{pmatrix}$$

At time $t = 1$, get $\left.\frac{d}{dt} P(r(t))\right|_{t=1} = \begin{pmatrix} 3-3 \\ 2+1 \\ 1-6 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}$

$$= -1$$

10. Determine all saddle points of the function

$$f(x,y) = 2x^3 - 6x + 3xy^2.$$

$(0, \sqrt{2})$ & $(0, -\sqrt{2})$

$f(x,y) = 2x^3 - 6x + 3xy^2 \Rightarrow f_x(x,y) = 6x^2 - 6 + 3y^2$

$f_y(x,y) = 6xy$

For critical points, set $f_x(x,y) = 0 = f_y(x,y)$

get $\begin{cases} 6x^2 - 6 + 3y^2 = 0 & (1) \\ 6xy = 0 & (2) \end{cases}$

If $x = 0$, then (2) holds, so for (1) to hold, must have $-6 + 3y^2 = 0$

ie $y^2 = 2$ so $y = \pm\sqrt{2}$

If $y = 0$, then (2) holds, so for (1) to hold, must have $6x^2 - 6 = 0$

ie $x^2 = 1$ so $x = \pm 1$

(If $x \neq 0$ and $y \neq 0$ then (2) fails)

Hence all critical points are $(0, \sqrt{2})$, $(0, -\sqrt{2})$, $(1, 0)$, $(-1, 0)$

Compute Hessian matrix $\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} 12x & 6y \\ 6y & 6x \end{pmatrix}$

	Hessian	det Hessian	type of critical point
eval at $(0, \sqrt{2})$, get	$\begin{pmatrix} 0 & 6\sqrt{2} \\ 6\sqrt{2} & 0 \end{pmatrix}$	-ve	saddle
$(0, -\sqrt{2})$	$\begin{pmatrix} 0 & -6\sqrt{2} \\ 6\sqrt{2} & 0 \end{pmatrix}$	-ve	saddle
$(1, 0)$	$\begin{pmatrix} 12 & 0 \\ 0 & 6 \end{pmatrix}$	+ve	local min
$(-1, 0)$	$\begin{pmatrix} -12 & 0 \\ 0 & -6 \end{pmatrix}$	+ve	local max